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RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

**BookVerse Retail Business
Intelligence & Sales Analytics**

About Me!

Geethika Dasari(Individual Project)

- **Background:** B.Tech in IT
- **Why Data Analytics:** To turn business data into actionable insights.
- **Work experience:** Fresher
- **LinkedIn:** [Link](#)
- **GitHub:** [Link](#)
- **Project role:** Data Analyst & Buisness Analyst — database design, SQL analysis, insights and recommendations.

Project Introduction

BookVerse Retail Pvt. Ltd. is a growing online and offline bookstore specializing in books across multiple genres, including Fiction, Fantasy, Biography, Mystery, Romance, Science Fiction, and Non-Fiction.

The company manages three core business operations:

- Book Catalog Management
- Customer Management
- Order & Sales Management

As the business continues to expand, management wants to use historical transactional data to understand **sales performance, customer purchasing behaviour, product demand, and inventory availability.**

Why this project ?

- Identify books and genres that contribute most to revenue.
- Understand which customers are most active and valuable.
- Measure product demand using sales quantity and revenue.
- Detect low-stock and excess-stock situations before procurement decisions.
- Convert raw transactional data into clear business actions using SQL.

Project Objectives

- Build the BookVerse relational database in MySQL.
- Validate records, missing values, duplicates, keys and relationships.
- Explore overall sales, customer and product performance.
- Analyse customer purchasing behaviour across countries and repeat orders.
- Evaluate inventory position and identify sales–inventory opportunities.
- Provide practical recommendations for promotions, retention and replenishment

Dataset Information

The SQL project contains three core relational tables.

BOOKS

496 records

Catalog, pricing and inventory information.

CUSTOMERS

500 records

Customer identity and geographic information.

ORDERS

495 records

Transactional sales and order information.

Books: Book_ID, Title, Author, Genre, Published_Year, Price, Stock.

Customers: Customer_ID, Name, Email, Phone, City, Country.

Orders: Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount.

Data Quality & Validation

BOOKS

496 distinct

Primary key uniqueness checked; null and duplicate checks performed.

CUSTOMERS

500 distinct

Customer IDs checked for uniqueness; null and duplicate checks performed.

ORDERS

495 distinct

Order IDs checked for uniqueness; null and duplicate checks performed.

The SQL script documents no null values in the checked columns and no duplicate records in the three tables.

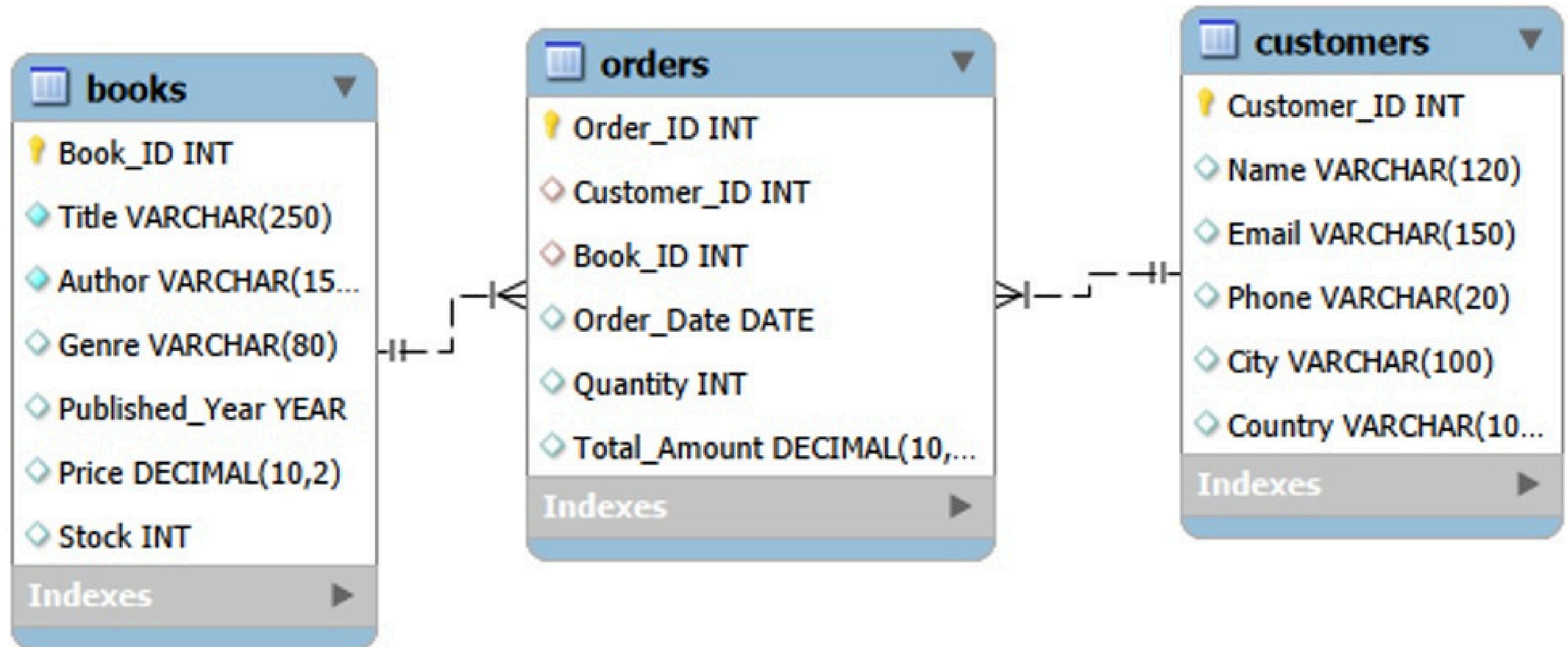
Foreign-key consistency was tested by checking Orders against Customers and Books.

A recursive query was also used to identify missing Book_ID values in the expected ID range.

Table Structure Explanation

Table	Key columns	Purpose
Books	Book_ID (PK)	Master data for products and inventory.
Customers	Customer_ID (PK)	Customer master and geography.
Orders	Order_ID (PK); Customer_ID, Book_ID (FK)	Transaction fact table connecting customers and books.

ER Diagram



Problem Statement

How can BookVerse use its transactional data to improve sales, customer retention, product performance and inventory decisions?

- Which books and genres generate the most revenue?
- Which customers contribute the most to sales?
- How frequently do customers purchase?
- Which books are selling the most?
- Which genres have high inventory value?
- Which books may require inventory attention?

SQL queries

- Top 5 Books by Revenue
- Revenue by Genre
- Top Customers by Spending
- Top Books by Quantity Sold
- High Sales + Low Stock
- High Stock + Low Sales

1. Which books generate the highest revenue?

```
SELECT
  b.book_id,
  b.title,
  SUM(o.total_amount) AS total_revenue
FROM books b
JOIN orders o
ON b.book_id = o.book_id
GROUP BY b.book_id, b.title
ORDER BY total_revenue DESC
LIMIT 5;
```

book_id	title	total_revenue
120	Integrated secondary access	1104.69
142	Multi-tiered responsive parallelism	1077.12
119	Switchable modular moratorium	1047.48
471	Cross-platform next generation website	952.98
422	Grass-roots systematic moderator	872.29

Business Insight: Identifies the major revenue-generating books so BookVerse can prioritize them for promotions and maintain sufficient stock.

2. Which genres generate the highest revenue?

```
SELECT
  b.genre,
  SUM(o.total_amount) AS total_rev
FROM books b
JOIN orders o
  ON b.book_id = o.book_id
GROUP BY b.genre
ORDER BY total_rev DESC;
```

	genre	total_rev
▶	Romance	12907.58
	Mystery	12394.73
	Science Fiction	11770.51
	Fantasy	11238.38
	Non-Fiction	11030.51

Business Insight: Shows which genres contribute most to revenue. Romance is the highest-revenue genre, making high-performing genres suitable for greater marketing focus.

3. Which customers contribute the most to sales?

```
SELECT
  c.customer_id,
  SUM(o.total_amount) AS high_amount
FROM customers c
JOIN orders o
  ON c.customer_id = o.customer_id
GROUP BY c.customer_id
ORDER BY high_amount DESC
LIMIT 5;
```

	customer_id	high_amount
▶	457	1398.90
	174	1080.95
	364	1052.27
	405	991.00
	386	986.30

Business Insight: Identifies high-value customers who contribute significantly to revenue. BookVerse can focus on customer retention through loyalty benefits and personalized offers.

4. Which books have the highest customer demand based on units sold?

```
SELECT
  b.book_id,
  b.title,
  SUM(o.quantity) AS sale_quantity
FROM books b
JOIN orders o
ON b.book_id = o.book_id
GROUP BY b.book_id, b.title
ORDER BY sale_quantity DESC
LIMIT 5;
```

	book_id	title	sale_quantity
►	73	Realigned multi-tasking installation	28
	31	Implemented encompassing conglomeration	27
	157	Advanced didactic time-frame	24
	273	Devolved zero administration process improvem...	24
	288	Re-contextualized real-time Graphic Interface	23

Business Insight: Identifies the books with the strongest customer demand based on units sold. These books should receive priority for inventory replenishment.

5. Which books have relatively high demand but low available stock?

```
SELECT
  b.book_id,
  b.title,
  SUM(o.quantity) AS stock,
  b.stock
FROM books b
JOIN orders o
  ON b.book_id = o.book_id
GROUP BY b.book_id, b.title, b.stock
HAVING SUM(o.quantity) > 10 AND b.stock <
20
ORDER BY stock
DESC;
```

	book_id	title	stock
▶	2	Persevering reciprocal knowledge user	19
	4	Customizable 24hour product	8
	6	Advanced encompassing implementation	2
	9	Optimized interactive challenge	70
	12	Polarized optimal array	63

Business Insight: Identifies the books with the strongest customer demand based on units sold. These books should receive priority for inventory replenishment.

6. Which books have relatively high stock but low demand?

```
SELECT
  b.book_id,
  b.title,
  b.stock,
  SUM(o.quantity) AS total_sold
FROM books b
JOIN orders o
  ON b.book_id = o.book_id
GROUP BY b.book_id, b.title, b.stock
HAVING b.stock > 20
      AND SUM(o.quantity) < 10
ORDER BY b.stock DESC;
```

	book_id	title	stock	total_sold
▶	1	Configurable modular throughput	100	3
	29	Quality-focused multi-tasking challenge	100	4
	475	Stand-alone zero administration emulation	100	1
	254	Team-oriented bandwidth-monitored project	100	4
	13	Adaptive 5thgeneration orchestration	99	9

Business Insight: Identifies books with excess inventory and weak sales. BookVerse can use discounts, bundles, or promotions to improve sales and reduce excess stock.

Insights — Sales Performance

- A small group of books contributes disproportionately to revenue; these should receive priority in promotions and stock planning.
- Top-selling books based on quantity indicate strong customer demand and should be replenished proactively.
- Romance is the highest-revenue genre in the documented analysis.
- Sheena Harris is the highest-revenue author in the documented analysis.
- Monthly and yearly sales analysis can be used to identify peak periods and support planning.

Insights — Customer & Product Performance

CUSTOMER BEHAVIOUR

- Compare customers by order count and total spending.
- Identify repeat buyers and high-value customers.
- Compare customer activity across countries.
- Use loyalty rewards and personalized offers for valuable customers.

PRODUCT / GENRE PERFORMANCE

- Track high-quantity and high-revenue books.
- Maintain wider selection in high-volume genres.
- Use promotions for low-selling books.
- Review books with zero recorded orders before future procurement.

Business Recommendations

- Replenish high-selling books with lowstock before the next procurement cycle.
- Prioritize high-revenue books and high-performing genres in promotions.
- Introduce loyalty rewards and personalized offers for frequent and high-spending customers.
- Use discounts or bundles to reduce high-stock, low-sales inventory.
- Review never-ordered books and reduce future procurement if demand remains weak.
- Target high-activity countries with focused marketing campaigns.

Conclusion, Challenges & Learning

Key Conclusion

BookVerse can use SQL-driven analysis to connect sales, customer behaviour and inventory decisions.

The project highlights clear opportunities around top revenue books, customer retention, demand-based replenishment and excess-stock reduction.

Challenges / Learning

Working with relational joins across three tables.

Validating data quality before analysis.

Translating business questions into measurable SQL queries.

Turning query outputs into practical business recommendations.

THANK YOU

